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Enhanced emission intensity of Ce^{3+} ions in Li_2O–B_2O_3–Gd_2O_3 scintillating glasses by adding carbon and Si_3N_4 agent

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ABSTRACT

Ce^{3+}-doped Li_2O–B_2O_3–Gd_2O_3 scintillating glasses containing neutron-capture elements were synthesized by melt quenching method in air. More attention is paid on the improved optical properties induced by adding carbon and Si_3N_4 as a reducing agent. The spectral results show that energy transfer efficiency from Gd^{3+} to Ce^{3+} ions reaches approximately 95%. With the incorporation of carbon and Si_3N_4 into the designed glass, the enhanced emission of Ce^{3+} ions can be further increased by a factor of about 3 and 4.5, respectively. The investigated Ce^{3+}-doped Li_2O–B_2O_3–Gd_2O_3 scintillating glass is characterized with the broad band centered at 360 nm and a lifetime of about 36–38 ns, which will be potential application for neutron detection.

1. Introduction

Neutron diagnostics is an indispensable tool for both inertial confinement fusion (ICF) and magnetic confinement fusion (MCF) research to effectively probe and control high-energy fusion plasma [1–3]. The available technology for neutron detection is generally based on nuclear reaction, which needs high performance scintillating materials containing neutron-capture elements including ^{6}Li, ^{10}B, ^{155}Gd and ^{157}Gd [4,5]. In the 1960s, neutron detection using the fast Ce^{3+}-activated silicate and borosilicate glasses was proposed by Ginther et al. for the first time [6–8]. The temperature dependence on light yield and decay time of Ce^{3+}-activated SiO_2–Al_2O_3–MgO glass was subsequently investigated by Spowart et al. [9,10], even the slow Tb^{3+}-activated silicate glasses were also developed for thermal neutron detection [11,12]. In recent years, novel Ce^{3+} and Pr^{3+}-activated $20Al(PO)_3$–80LiF glasses are also applied for neutron diagnosis owing to the superfast Ce^{3+} and Pr^{3+} ions [2,3].

The $Li_2B_4O_7$ glass is one of the most important borate materials lying in its high photo-, and thermo-mechanical stability, and a wide spectral range of transparency even in the ultra-UV region [13]. Hence, the crystalline or amorphous $Li_2B_4O_7$ materials codoped with certain activators have been extensively utilized as a nonlinear optical material, thermoluminescence dosimetry and neutron detection aim [13–15]. To our best knowledge, the scintillating properties of Ce^{3+} ions in $Li_2B_4O_7$ glass have not been reported [16,17]. Therefore, in the present work, Ce^{3+}-activated $Li_2B_4O_7$ scintillating glasses are synthesized in air atmosphere by adding appropriate carbon and Si_3N_4 powder as a strong reducing agent for the first time [18]. The scintillating properties are comparatively investigated by the optical transmittance, photoluminescence, and X-ray excited luminescence (XEL) spectra.

2. Experimental methods

Scintillating glasses with the detailed compositions are listed in Table 1. Batches of about 13 g of raw materials were well ground in an agate mortar and melted in an alumina crucible at 1000 °C for about 30 min in air. The homogeneous melts were quickly poured onto a preheated stainless steel mold. The quenched glasses are finally annealed at 400 °C for 4 h followed by cooling down naturally to room temperature.

Optical transmittance spectra were taken on a Perkin Elmer Lambda 750S UV/VIS spectrometer in the 200–800 nm region. Excitation and emission spectra were recorded on a Hitachi F-7000 fluorescence spectrophotometer, equipped with a 150 W xenon lamp as the excitation source. Luminescence decay curves of Ce^{3+} ions were performed on an Edinburgh FLS980 spectrometer with excitation photons from a 150 W nF900 ns flash lamp with a pulse width of 1 ns and pulse repetition rate of 40–100 kHz. The radioluminescence light (irradiated by Cu K_α X-rays operating at 30 kV and 3 mA) were coupled to the same FLS980 spectrometer via optical lens and fibers. All measurements were carried out at room temperature.

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Table 1
Glass composition (unit: g).

	Li_2CO_3	H_3BO_3	Gd_2O_3	CeO_2	CeF_3	Carbon	Si_3N_4	Description
No. 1	2.6001	8.7031	1.4948	0	0	0	0	Host
No. 2	2.6001	8.7031	1.4948	0.0473	0	0	0	CeO_2-doped
No. 3	2.6001	8.7031	1.4948	0	0.0542	0	0	CeF_3-doped
No. 4	2.6001	8.7031	1.4948	0	0.0542	0.0380	0	Carbon-reduced
No. 5	2.6001	8.7031	1.4948	0	0.0542	0	0.0120	Si_3N_4-reduced

3. Results and discussion

3.1. Optical transmittance spectra

Fig. 1 shows the optical transmittance spectra of Ce^{3+}-doped scintillating glasses in 200–800 nm region. The optical absorption peaks at 276 and 313 nm assigned to the Gd^{3+} $^8S_{7/2} \rightarrow {}^6I_J$, ^6P_J transitions are clearly observed in the host glass (No. 1), they completely vanishes with the incorporation of Ce^{3+} ions in view of the noticeable red shift of the cut-off edges. This red shift gets weaker in the order of No. 2, No. 3, No. 4 and No. 5 glasses. According to our previous results, the position of cut-off edge is determined by the concentration of Ce^{3+} ions in the same host glasses [18,19]. Generally, the higher the concentration of Ce^{3+} ions in glass, the shorter wavelength the cut-off edge of glass will be. The red shift of cut-off edge in CeO_2-doped glass (No. 2) is largest, which implies the lowest concentration of Ce^{3+} ions. While the shortest one shows in Si_3N_4-reduced glass (No. 5), which indicates the highest concentration of Ce^{3+} ions is. This can also be illustrated by the digital photographs of the investigated glasses, where the color of the Ce^{3+}-doped glasses is observed to change from slight yellow to colorless with the incorporation of either carbon or Si_3N_4 reducing powder, as shown in the inset of Fig. 1.

In addition, the linear transmittance coefficient of all Ce^{3+}-doped Li_2O–B_2O_3–Gd_2O_3 glasses exceeds 80%, even reaches 90%, in the 425–800 nm region. The high optical transmittance is of significance in the higher light yield of a scintillator.

3.2. Excitation and emission spectra

Fig. 2 shows the excitation ($\lambda_{em} = 358$ nm) and emission ($\lambda_{ex} = 276$ and 308 nm) spectra of Ce^{3+}-undoped (No. 1) and Ce^{3+}-doped (No. 2–No. 5) glasses. As shown in the excitation spectra of Fig. 2(a), two groups of broad bands in 250–300 and 300–400 nm regions are ascribed to the 4f–$5d_{3,4,5}$ and 4f–$5d_{1,2}$ transition, respectively [18]. Under direct excitation at the 4f–$5d_{1,2}$, a typical emission $5d^1$-$^2F_{5/2,7/2}$ Ce^{3+} emission extending from 300 to 450 nm is observed, as shown in the emission spectra of Fig. 2(b) and (c). The following results can be easily found: (1) the emission intensity of Ce^{3+} ions in CeF_3-doped glass is 8–10 times higher than that CeO_2-doped one; (2) the emission intensity of CeF_3-doped glass can be further increased by about 3 and 4.5 times by adding carbon and Si_3N_4 powder, respectively; (3) and the most strong emission intensity of Ce^{3+} ions is the Si_3N_4-reduced glass (No. 5).

The obtained spectral results indicate that the reduction of cerium ions from trivalent cerium (Ce^{4+}) to its trivalent state (Ce^{3+}) can be effectively controlled in air atmosphere only by adding amounts of reducing carbon and Si_3N_4 agent.

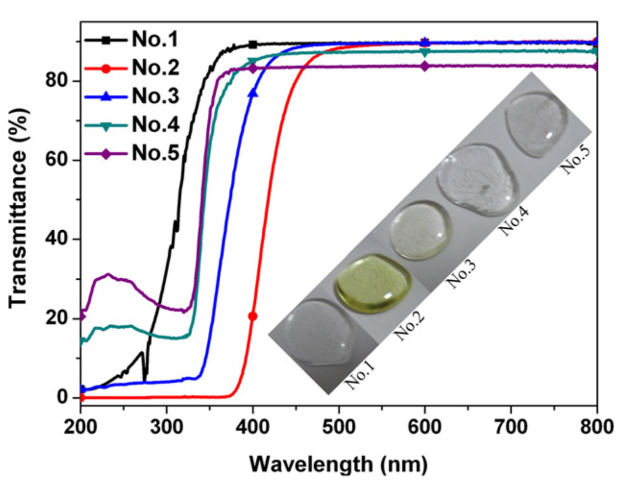

Fig. 1. Optical transmittance spectra. Inset shows the digital photographs.

3.3. Energy transfer from Gd^{3+} to Ce^{3+} ions

As emphasized in the introduction section, neutron-capture elements ^6Li, ^{10}B, ^{155}Gd and ^{157}Gd with high capture cross-section facilitate for higher probability of capturing thermal neutrons. Another crucible role of Gd^{3+} ions in Li_2O–B_2O_3–Gd_2O_3 scintillating glass may be its possible energy transfer from Gd^{3+} ions to the incorporated Ce^{3+} ions. We have noticed that the emission intensity of Gd^{3+} ions in Fig. 2(b) decreases dramatically by a factor of about 4–10 with the incorporation of Ce^{3+} ions into the Li_2O–B_2O_3–Gd_2O_3 glass, which verifies this efficient energy transfer process. To well understand it, the luminescence decay curves of Gd^{3+} ions ($\lambda_{ex} = 276$ nm, $\lambda_{em} = 313$ nm) were also measured and shown in Fig. 3. Without Ce^{3+}-codoping, the luminescence decay curve of Gd^{3+} ions is perfectly fitted with a single-exponential function with a lifetime of about 4401.07 μs, as seen in Fig. 3(b). However, as shown in Fig. 2(a), the luminescence decay curves of Gd^{3+} ions significantly deviate from the single exponential rule with the codoping of Ce^{3+} ions, which suggests that Ce^{3+}-codoping modifies the luminescence dynamics of the Gd^{3+} ions [20]. It is found that all the non-exponential curves can be well fitted by two exponentials. Namely, the mean lifetimes are evaluated by a sum of two exponential functions:

$$I(t) = A_1 \exp\left(-\frac{t}{\tau_1}\right) + A_2 \exp\left(-\frac{t}{\tau_2}\right) \tag{1}$$

where $I(t)$ is the emission intensity of Gd^{3+} ions at a given time, and A_1 and A_2 are constants, τ_1 and τ_2 are the short- and long-decay components, respectively. Thus, the mean decay time τ can be obtained by [20]:

$$\tau = \frac{A_1\tau_1^2 + A_2\tau_2^2}{A_1\tau_1 + A_2\tau_2}. \tag{2}$$

The mean lifetime values were evaluated to be about 173.88, 239.96, 233.32 and 155.43 μs, respectively. It is clearly observed that the decay of Gd^{3+} ions becomes more fast with the incorporated Ce^{3+} concentration, due to the efficient energy transfer from Gd^{3+} to Ce^{3+} ions.

A simple experimental formula can be used to estimate the energy-transfer efficiency from Gd^{3+} to Ce^{3+} ($\eta_{Gd \rightarrow Ce}$) by the following equation [21].

$$\eta_{Gd \rightarrow Ce} = 1 - \frac{\tau}{\tau_0} \tag{3}$$

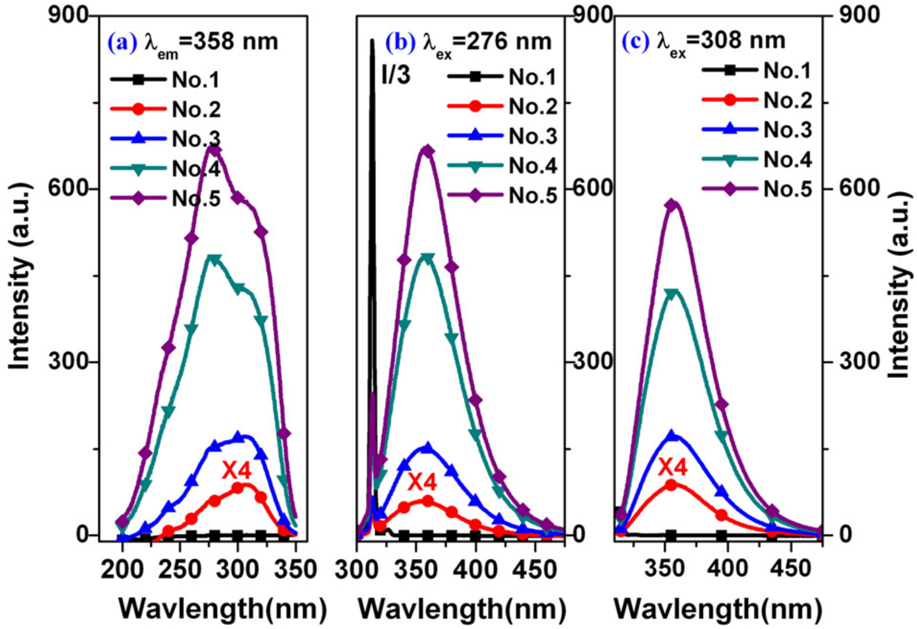

Fig. 2. Excitation (a) spectra by monitoring 358 nm, emission spectra excited by 276 nm (b) and 308 nm (c) light, respectively. The relative intensity of No. 2 glass is plotted four times, while the Gd^{3+} emission intensity of No. 1 glass is divided by 3 for clarity.

where τ_0 and τ are the Gd^{3+} donor lifetime in the absence and presence of a Ce^{3+} acceptor, respectively. Taken the evaluated lifetimes of Gd^{3+} ions into Eq. (3), the calculated values of $\eta_{Gd \rightarrow Ce}$ are 96.05%, 94.55%, 94.70% and 96.47%, respectively. The energy transfer efficiency is high enough to be close to the unit, which is in good accordance with that of the emission spectra of Fig. 2(b).

3.4. Luminescence decay curves

The lifetime, one of the most important parameters of a scintillator, should be taken into full consideration in practice. The luminescence decay curves (λ_{ex} = 308 nm, λ_{em} = 358 nm) of Ce^{3+} ions in Li_2O–B_2O_3–Gd_2O_3 glasses are shown in Fig. 4. It is clear that all the luminescence decay curves of Ce^{3+} ions significantly deviate from the single exponential rule. Interestingly, all the non-exponential curves can be well fitted by two exponentials. Therefore, the mean lifetimes are also evaluated by the Eq. (2). The fitted mean lifetimes for the No. 2, No. 3, No. 4 and No. 5 glasses are 20.77, 31.55, 38.06 ns and 36.81 ns, respectively. The increasing lifetimes of Ce^{3+} ions is related to the stronger emission intensity, which is well in line with the emission spectra, as shown in Fig. 2(b) and (c).

3.5. XEL spectra

The XEL spectra of Ce^{3+}-doped Li_2O–B_2O_3–Gd_2O_3 glasses under X-ray (Cu target, 30 kV and 3 mA) excitation are displayed in Fig. 5. The XEL intensity increases in the order of No. 2, No. 3, No. 4 and No. 5 glasses, the law is in line with the emission spectra of Fig. 2(b) and (c). To our surprise, the XEL intensity of Gd^{3+} ions in all glasses, as well as Ce^{3+} ions in No. 2 glass, is almost quenched. Additionally, a slight blue shift of Ce^{3+} emission is detected when compared with No. 3, No. 4 and No. 5 glasses. The reabsorption of Ce^{4+} ions in

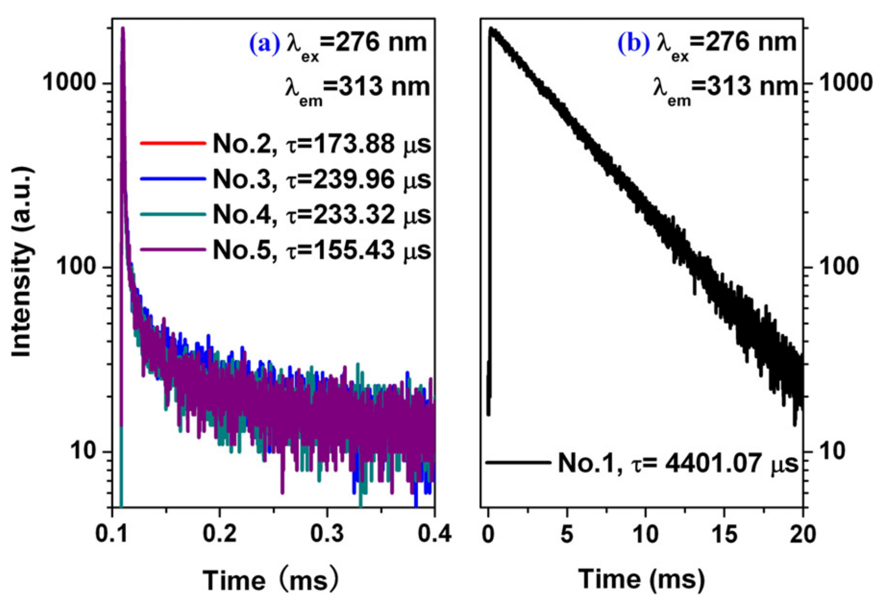

Fig. 3. Luminescence decay curves of Gd^{3+} ions in Li_2O–B_2O_3–Gd_2O_3 glasses doped with Ce^{3+} (a) and without Ce^{3+} ions (b).

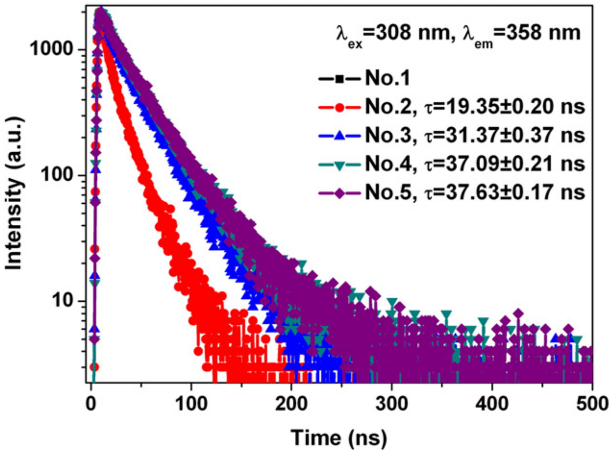

Fig. 4. Luminescence decay curves of Ce^{3+} ions in $Li_2O–B_2O_3–Gd_2O_3$ glasses.

$Li_2O–B_2O_3–Gd_2O_3$ glasses may be responsible for the slight blue shift in the XEL spectra, as proved in the optical transmittance spectra of Fig. 1. Another plausible fact may be attributed to the different excitation mechanism between ultraviolet and X-ray light. The physical principle of a scintillator is well known to be dominated by three successive physics processes: Transfer of the energy from the electron–hole pairs initially created by incident radiation to a luminescence center [22]. Therefore, the efficiency of a scintillator is determined by $Y = N_{eh}SQ$, where N_{eh} is the conversion efficiency expressed as a number of electron–hole pairs or excitons, S is the probability of transfer to emitting centers, and Q is the luminescence quantum yield. Therefore, the Ce^{3+} ions in the present scintillating glasses should be indirectly excited during X-ray irradiation. However, ultraviolet light excitation process can be regarded as the direct excitation of Ce^{3+} ions in the investigated scintillating glasses.

4. Conclusions

The energy transfer efficiency from Gd^{3+} to Ce^{3+} ions in Ce^{3+}-activated $Li_2O–B_2O_3–Gd_2O_3$ scintillating glasses reaches approximately 95%. The emission intensity of CeF_3-doped glass is much stronger than that of CeO_2-doped one by a factor of 8–10. Upon incorporation of carbon and Si_3N_4 into the CeF_3-doped glass, the emission intensity of Ce^{3+} ions can be further increased by a factor of about 3 and 4.5, respectively. The spectral results show that Ce^{3+}-doped $Li_2O–B_2O_3–Gd_2O_3$ glasses with a broad band centered at 360 nm and a lifetime of about 36–38 ns, which will be potential application for neutron detection.

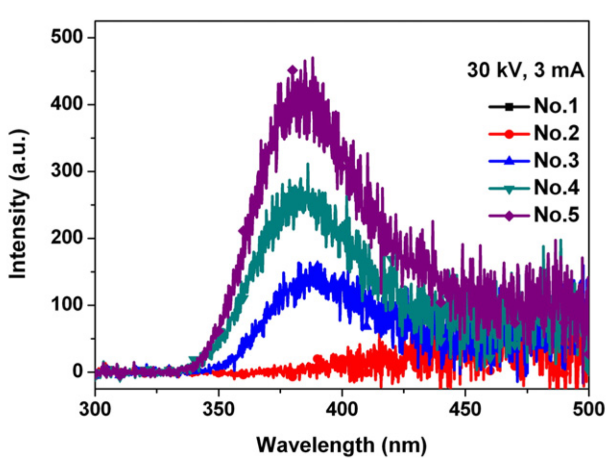

Fig. 5. XEL spectra.

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References

[1] D.A. Haas, M. Bliss, S.M. Bowyer, J.D. Kephart, M.J. Schweiger, L.E. Smith, Actinide-loaded glass scintillators for fast neutron detection, Nucl. Instrum. Methods Phys. Res. A 652 (2011) 421–423.
[2] Y. Arikawa, K. Yamanoi, T. Nakazato, E.S. Estacio, T. Shimizu, N. Sarukura, M. Nakai, T. Norimatsu, H. Azechi, T. Murata, S. Fujino, H. Yoshida, K. Kamada, Y. Usuki, T. Suyama, A. Yoshikawa, N. Sato, H. Kan, Pr^{3+}-doped fluoro-oxide lithium glass as scintillator for unclear fusion diagnostics, Rev. Sci. Instrum. 80 (2009) 113505.
[3] K. Watanabe, Y. Arikawa, K. Yamano, M. Cadatal-Raduban, T. Nagai, M. Kouno, K. Sakai, T. Nazakato, T. Shimizu, N. Sarukura, M. Nakai, T. Norimatsu, H. Azechi, A. Yoshikawa, T. Murata, S. Fujino, H. Yoshida, N. Izumi, N. Satoh, H. Kan, Pr or Ce-doped, fast-response and low-afterglow cross-section-enhanced scintillator with ^{6}Li for down-scattered neutron originated from laser fusion, J. Cryst. Growth 362 (2013) 288–290.
[4] B.M. Fisher, J.N. Abdurashitov, K.J. Coakley, V.N. Gavrin, D.M. Gilliam, J.S. Nico, A.A. Shikhin, A.K. Thompson, D.F. Vecchia, V.E. Yants, Fast neutron detection with ^{6}Li-loaded liquid scintillator, Nucl. Instrum. Methods Phys. Res. A 646 (2011) 126–134.
[5] J.P. Chaminade, O. Viraphong, F. Guillen, C. Fouassier, B. Czirr, Crystal growth and optical properties of new neutron detectors Ce^{3+}:$Li_6R(BO_3)_3$ (R = Gd, Y), IEEE Trans. Nucl. Sci. 48 (2001) 1158–1161.
[6] F.W.K. Firk, G.G. Slaughter, R.J. Ginther, An improved Li^6-loaded glass scintillator for neutron detection, Nucl. Instrum. Methods 13 (1961) 313–316.
[7] R.J. Ginther, New cerium activated scintillating glasses, IRE Trans. Nucl. Sci. 7 (1960) 28–31.
[8] L.M. Bollinger, G.E. Thomas, R.J. Ginther, Neutron detection with glass scintillator, Nucl. Instrum. Methods 17 (1962) 97–116.
[9] A.R. Spowart, Neutron scintillating glasses: part I: activation by external charged particles and thermal neutrons, Nucl. Instrum. Methods 135 (1976) 441–453.
[10] A.R. Spowart, Neutron scintillating glasses: part II: the effects of temperature on pulse height and conductivity, Nucl. Instrum. Methods 140 (1977) 19–28.
[11] G.B. Spector, T. McCollum, A.R. Spowart, Advances in terbium-doped, lithium-loaded scintillator glass development, Nucl. Instrum. Methods Phys. Res. A 326 (1993) 526–530.
[12] G. Zanella, R. Zannoni, R. Dall'Igna, P. Polato, M. Bettinelli, Development of a terbium-lithium glass for slow neutron detection, Nucl. Instrum. Methods Phys. Res. A 359 (1995) 547–550.
[13] S. Kar, S. Bairagi, C. Debnath, S. Verma, K.S. Bartwal, Thermoluminesence studies on gamma-irradiation Mn:$Li_2B_4O_7$ single crystals, Appl. Phys. Lett. 101 (2010) 071904.
[14] V.T. Adamiv, Ya V. Burak, I.V. Kityk, J. Kasperczk, R. Smok, M. Czerwinski, Nonlinear optical properties of $Li_2B_4O_7$ single crystals doped with potassium and silver, Opt. Mater. 8 (1997) 207–213.
[15] A. Majchrowski, I.V. Kityk, E. Mandowska, A. Mandowski, J. Ebothe, T. Lukasiewicz, Several features of emission spectra of Pr^{3+} ions incorporated into $Li_2B_4O_7$ glass matrixes, J. Appl. Phys. 100 (2006) 053101.
[16] B.I. Zadneprovski, N.V. Eremin, A.A. Paskhalov, New inorganic scintillators on the basis of LBO glass for neutron registration, Funct. Mater. 12 (2005) 261–267.
[17] P.P. Puga, G.D. Puga, K.P. Popovich, V.A. Kelman, V.N. Krasylynec, I.I. Turok, M.V. Prymak, P.S. Danyliuk, Optical absorption and X-ray luminescence of glassy lithium tetraborate activated by terbium oxide, Glass Phys. Chem. 38 (2012) 190–195.
[18] X. Sun, Z. Ye, Y. Wu, P. Gao, R. Mao, Z. Zhang, J. Zhao, A simple and highly efficient method for synthesis of Ce^{3+}-activated borogermante scintillating glasses in air, J. Am. Ceram. Soc. 97 (2014) 3388–3391.
[19] X. Sun, L. Lin, Y. Wu, P. Gao, Z. Xiao, Enhanced Ce3+ Emission in $B_2O_3–GeO_2–Gd_2O_3$ scintillating glasses induced by melting temperature, Opt. Mater. Express 5 (2015) 920–925.
[20] X. Sun, Z. Ye, Z. Zhang, L. Liu, D. Chen, J. Zhao, Energy-transfer properties of dense Eu^{3+}/Tb^{3+}-coactivated oxyfluoride borogermante scintillating glasses, J. Am. Ceram. Soc. 98 (2015) 781–787.
[21] R. Reisfeld, E. Greenberg, R. Velapoldi, B. Barnet, Luminescence quantum efficiency of Gd and Tb in borate glasses and the mechanism of energy transfer between them, J. Chem. Phys. 56 (1972) 1698–1705.
[22] X. Sun, M. Gu, M. Zhang, S. Huang, Influence of CeO_2 on scintillating properties of Tb^{3+}-doped silicate glasses, J. Rare Earths 28 (2010) 340–344.